CSIT431 Introduction to Robotics/CSIT531 Robotics

Comprehensive Course Project

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Basic Information

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- **Course project:** **Development of a robot gripper system and a gesture recognition system to control it**

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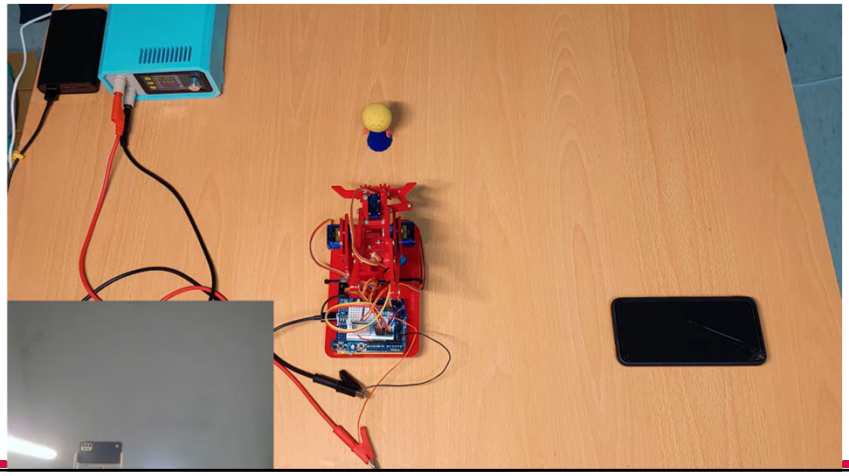


Basic Information

- The project takes 55% of the full grade.

Comprehensive Project	Competition Demonstration (60%)	55%
	Presentation (10%)	
	3D Model File (15%)	
	Code (15%)	

- Projects will be accomplished by groups and each group contains about 4 students.
- Each group will be graded based on the project implementation results (including ***on-site demonstration, presentation, 3D gripper design model file, Python code for hand gesture recognition, and Arduino code for Servo control***).

Basic Information

- The robot electronics integration, robot programming, motor controls, signal processing, computer vision, and system controls will be used in the project.
- The instructor will provide each group with all electronics parts for the robot gripper development.
- Free 3D printer can be found in the MSU MIX Lab: https://www.montclair.edu/entrepreneur/innovation-3d-printing/mix-lab/.
- Teamwork is very significant in the course project. In each project group, students should collaborate with and respect each other.

Schedule

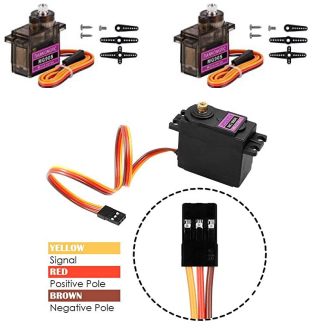
Robotics design and development: comprehensive project kickoff meeting and 3D printing basics
Robot gripper design and 3D printing
Robot gripper electronics integration
Computer vision design and development for robot control
Robot gripper testing
Comprehensive project competition and presentation

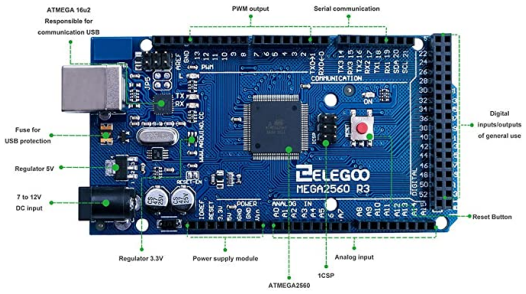
Find the dates on syllabus

Hardware and Software System

Hardware and Software System

- In this course project, students will employ **motors, Arduino MEGA2560 board, and 3D printers** as well as the fundamental knowledge of robotics delivered in this course, such as **robot electronics integration, robotics programming, signal processing, computer vision, and system controls**.





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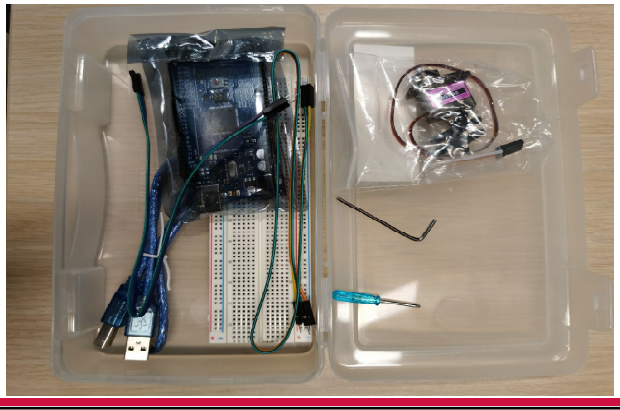
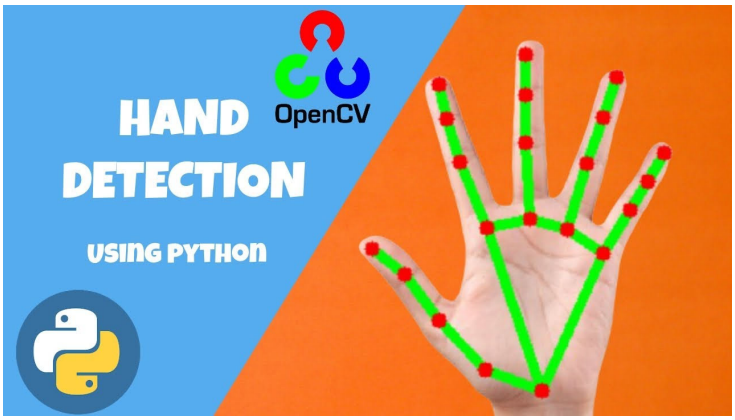
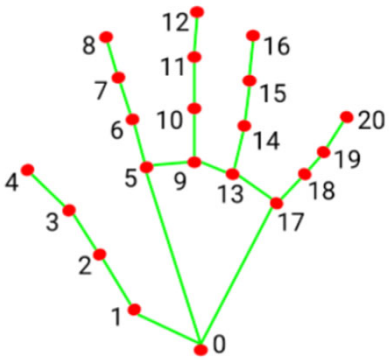


Hardware and Software System

- In this course project, students will employ **Python to develop a hand gesture recognition system to control the gripper to open or close to grasp objects**.





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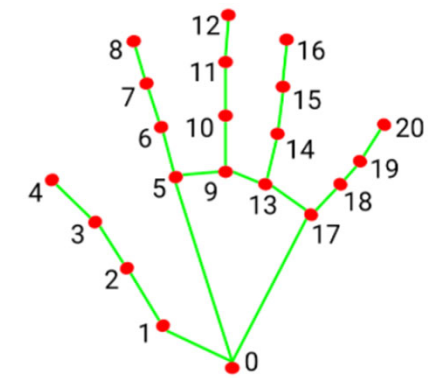

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3. THUMB_IP
4. THUMB_TIP
5. INDEX_FINGER_MCP
6. INDEX_FINGER_PIP
7. INDEX_FINGER_DIP
8. INDEX_FINGER_TIP
9. MIDDLE_FINGER_MCP
10. MIDDLE_FINGER_PIP
11. MIDDLE_FINGER_DIP
12. MIDDLE_FINGER_TIP
13. RING_FINGER_MCP
14. RING_FINGER_PIP
15. RING_FINGER_DIP
16. RING_FINGER_TIP
17. PINKY_MCP
18. PINKY_PIP
19. PINKY_DIP
20. PINKY_TIP

Grading Policy

Grading Policy

- The project takes 55% of the full grade.
- Each group will be graded based on the project implementation results (including ***on-site demonstration, presentation, 3D gripper design model file, Python code for hand gesture recognition, and Arduino code for Servo control***).

Comprehensive Project	Competition Demonstration (60%)
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Grading Policy

- **On-site demonstration** (Please see the competition section).
- **Presentation**: ***each group needs to do a 5 – 7 minutes of on-site presentation using PPT (5 – 10 pages) in the classroom on the final demonstration day***. Present and explain the design and development of your robot gripper system and control system using hand gesture recognition. Including but not limited to:
  - ✓ *How do you design and develop the gripper*?
  - ✓ *What are the strengths of your design*?
  - ✓ *How do you design and develop the hand gesture recognition and gripper control system*?
  - ✓ *What are the strengths of your robot gripper control system and user experience?*
  - ✓ *Limitations of your design and potential solutions to improve it?*
  - ✓ …

Grading Policy

- **Quality of code**: the organization, readability (comments), concision, and reusability of the codes.
- **3D Gripper Design Model File**: completeness, originality, complexity.
- **Submissions to CANVAS: Please upload your code, PPT (after your on-site presentation), and 3D gripper model file to CANVAS -> Comprehensive Project before Dec. 10, 2025.**
- **Each group only needs one copy of the submission by the group leader or one of the designated group members.**

Competition

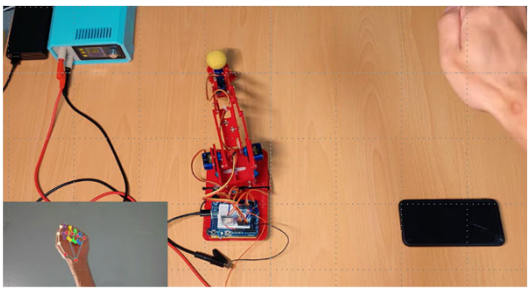
Competition

- **Time:** 5:30 pm - 8:00 pm, Tue., Dec. 2, 2025
- **Teams**: 8 teams
- **Location:** University Hall 1010

Competition

- **Task**

  Each group uses your developed hand gesture recognition system to control your robot gripper to grasp different shapes of objects from one box, then deliver and place them into another box.



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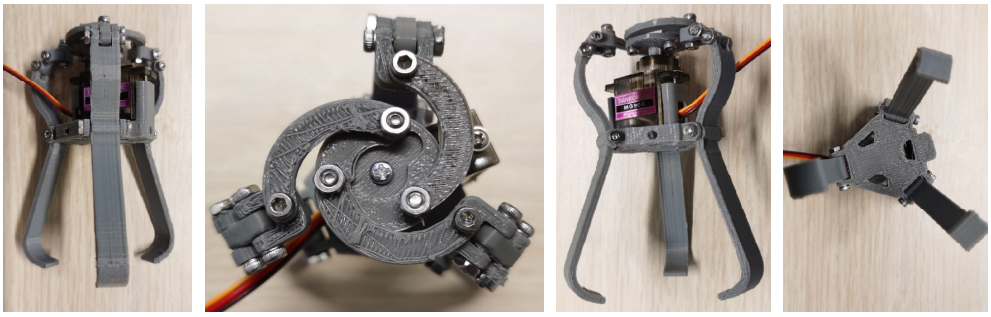

Competition

- **Rules**

  1) There are 20 objects with different shapes that need to be grasped.
  2) The distance between the two boxes is about 30 cm.
  3) Each object will be labeled with a corresponding score.
  4) Each team has three trials to grasp each object. If you successfully grasp an object at the first trial, you will automatically lose the two remaining trials for this object.
  5) In each grasping trial, your robot gripper should be controlled to open and close by your hand gesture recognition system.
  6) Each trial for each object grasping should be done within 10 seconds. If you cannot complete one trial within 10 seconds, please stop and start your next trial.

Competition

- **Rules**
  7) If the object **falls into the object holder** box from your gripper during the delivery process, no points for this trial.
  8) If the object **falls outside of the object holder box** from your gripper during the delivery process, you will get the corresponding points on the object. But 1 point will be deducted.
  9) If your gripper loses control, 2 points/loss will be deducted. 8 points at most will be deducted.
  10) Judges for each team will record the scores during the grasping process with a performance sheet.
  11) Each team's final score will be determined by the robot gripper's performance.

Competition

- **Performance sheet**

On-site demonstration performance sheet (60 points)											
Metrics	**Points**										
A: The grasped objects' points	***Object***	***Object 1***	***Object 2***	***Object 3***	***Object 4***	***Object 5***	***Object 6***	***Object 7***	***Object 8***	***Object 9***	***Object 10***
	Points										
	Object	***Object 11***	***Object 12***	***Object 13***	***Object 14***	***Object 15***	***Object 16***	***Object 17***	***Object 18***	***Object 19***	***Object 20***
	Points										
	Total Points for A										
B: Number of falls outside of the object holder box					***Total Points Deducted for B*** *(1 point will be deducted for each fall)*						
C: Number of gripper's lost controls					***Total Points Deducted for C*** *(2 points will be deducted for each loss)*						
Total points = A - B - C											

Presentation

Scoring Rubrics for Robotics Course Comprehensive Project Presentation

How to Do Your Project?

Teaching Assistant

- **Teaching Assistant: Xu Du (dux3@mail.montclair.edu)**
- Will be in charge of all activities in the course project

How to Do Your Project?

- Each group, at each class time, please schedule a group meeting to talk about your project plans and progress, work on your robot gripper, hand gesture recognition system, and control system design and development.
- After you complete the 3D model design for your gripper, please print it out using a 3D printer (our TA will show you how to print it).
- You can print it anytime. But you are suggested to print ALL parts and complete the gripper installation and testing **TWO WEEKS BEFORE** our final demonstration and presentation.

How to Do Your Project?

- When you integrate and assemble your robot gripper, please ask your TA for **screws** during our class time.
- **Attendance verification:** Your TA will do attendance verification for each group in each class. For any kind of reasonable absence, please let your TA know in advance by email.

Resources

Resources

- (Very important) Check and install ***Servo*** library in your Arduino IDE:

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- How to control Servo motor with Arduino: https://makersportal.com/blog/2020/3/14/arduino-servo-motor-control
- How to control Servo motor with Python and Arduino: https://github.com/jumejume1/python-arduino-basic/tree/main/servo-control

Resources

- Some robot gripper projects
- #1 EEZYbotARM 3D print: https://www.thingiverse.com/thing:2195839
- #2 Simple 3d printed 9g servo gripper 3D print: https://www.thingiverse.com/thing:2302957
- #3 Gripper robot arm 3D print: https://www.thingiverse.com/thing:2195839
- #4 Lightweight Gripper for FPV Drone 3D print: https://www.thingiverse.com/thing:2195839
- #5 Mini servo gripper Plate 3D print: https://www.thingiverse.com/thing:7109
- #6 Gripper robot arm 3D print: https://www.thingiverse.com/thing:2195839
- #7 Rack & Pinion Robotic Gripper Jaw 3D print: https://www.thingiverse.com/thing:2661755
- #8 Robot Gripper with three fingers for Chess Robot 3D print: https://www.thingiverse.com/thing:1322867
- #9 Parametric Robot Claw 3D print: https://www.thingiverse.com/thing:27468
- #10 Complet robotic arm (Bras robotique complet) 3D print: https://www.thingiverse.com/thing:2667301

Resources

- Array data processing with Arduino: https://www.arduino.cc/reference/en/language/variables/data-types/array/
- Arduino if statement: https://www.arduino.cc/reference/en/language/structure/control-structure/if/
- Arduino for loops: https://www.arduino.cc/reference/en/language/structure/control-structure/for/
- Arduino while loops: https://www.arduino.cc/reference/en/language/structure/control-structure/while/